

17-803 Empirical Methods

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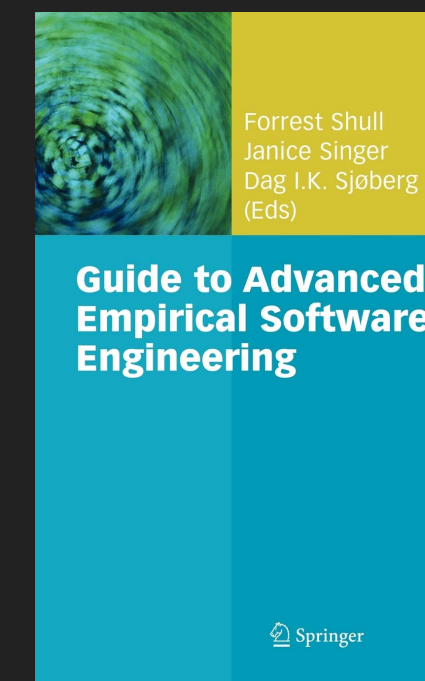
Designing Experiments (II)

Thursday, March 12, 2026

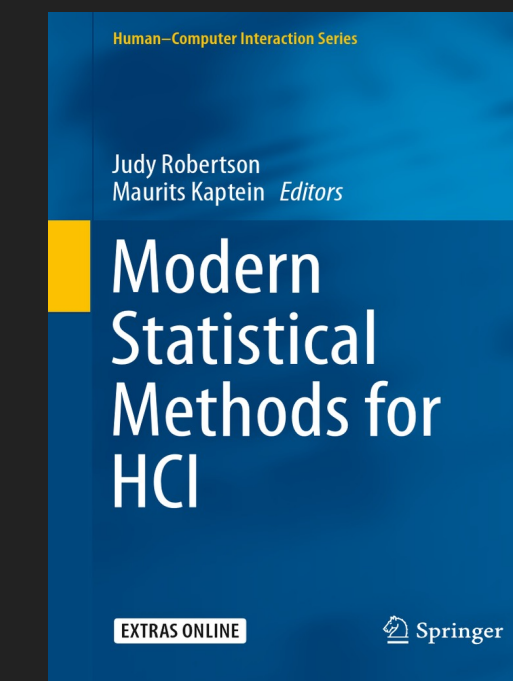
Readings



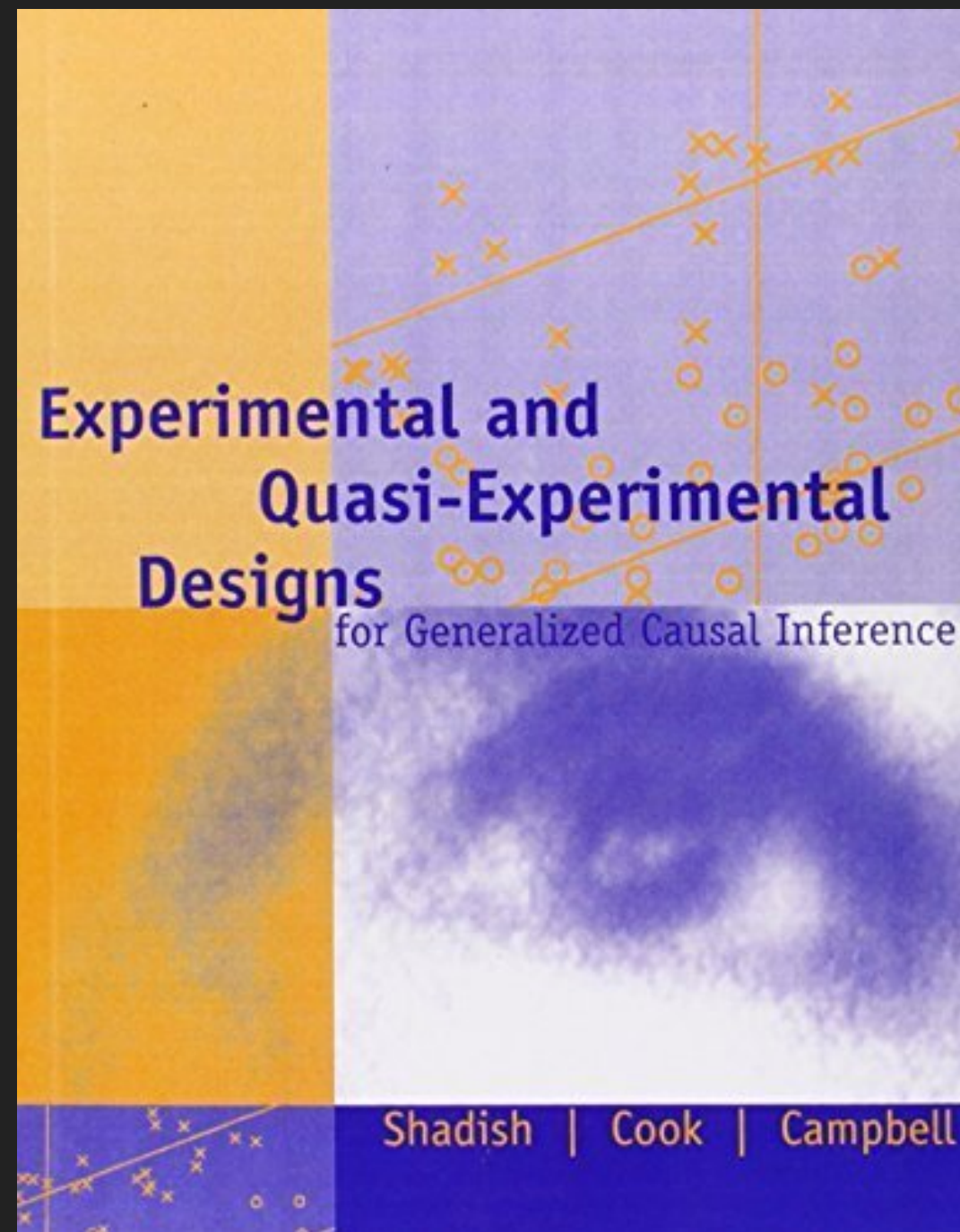
Ch 10 (Analysis and interpretation)



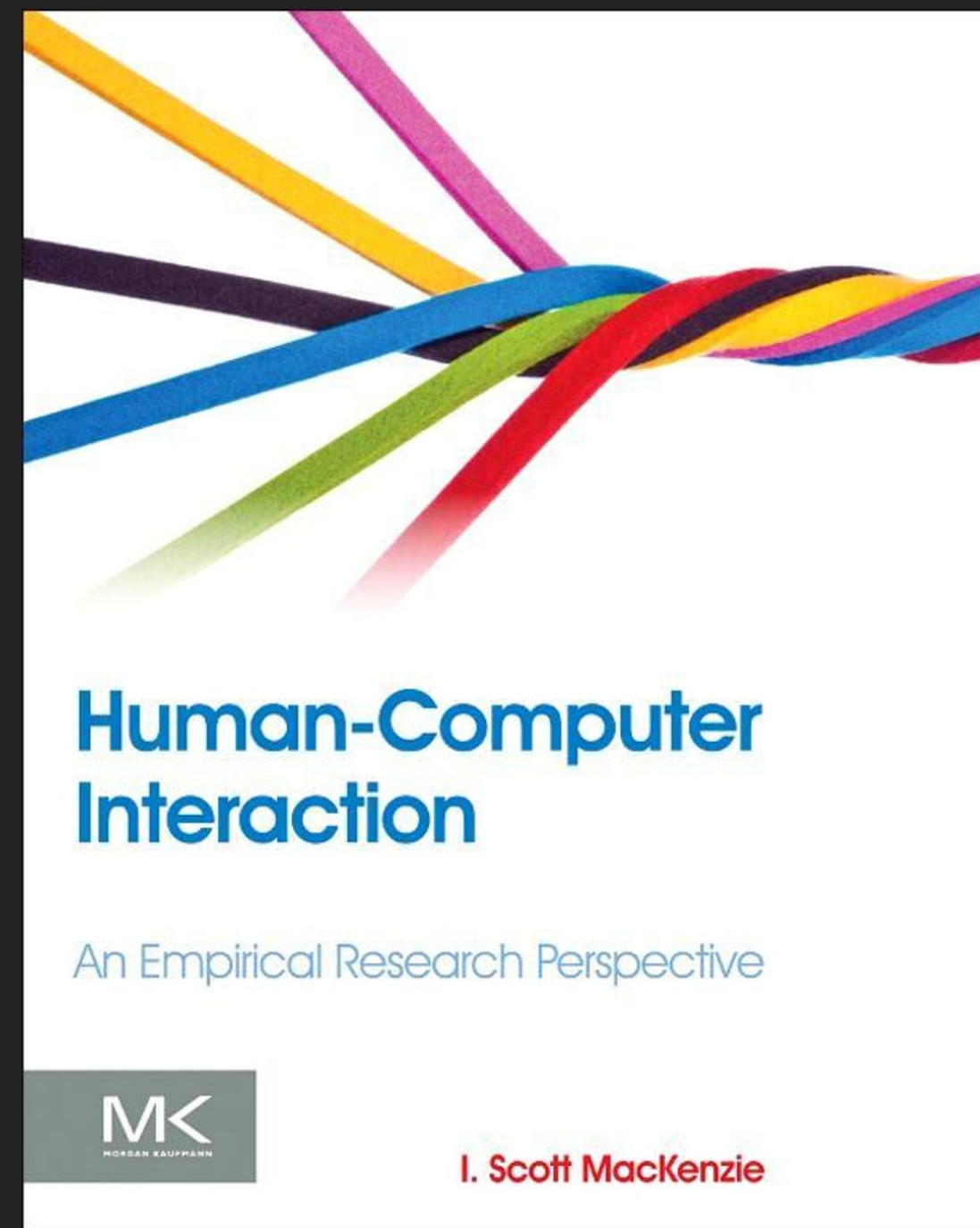
Ch 6 (Statistical methods and measurement)



Ch 5 (Effect sizes and power analysis)
Ch 13 (Fair statistical communication)
Ch 14 (Improving statistical practice)



Ch 1 (Experiments and causality)
Ch 2 & 3 (Validity)
Ch 8 (Randomized experiments)



Ch 5 (Designing HCI Exp.)
Ch 6 (Hypothesis testing)



Ch 3 (Experimental design)
Ch 4 (Statistical analysis)

Example paper presentations

NeurIPS (Conference on Neural Information Processing Systems) Experiment

▶ Setup

- ▶ Organizers split the program committee down the middle
- ▶ Most submitted papers were assigned to a single side
- ▶ 10% of submissions (166) were reviewed by both halves of the committee

▶ Results

- ▶ "most papers [57%] at NeurIPS would be rejected if one reran the conference review process (with a 95% confidence interval of 40-75%)"

Investigating more than one independent variable

Basic X vs C

R	X	O
R		O

Basic X_A vs X_B

R	X_A	O
R	X_B	O

Basic X_A vs X_B vs C

R	X_A	O
R	X_B	O
R		O

Pretest-posttest

R	O	X	O
R	O		O

Alternative Xs with pretest

R	O	X_A	O
R	O	X_B	O

Factorial

R	X_{A1B1}	O
R	X_{A1B2}	O
R	X_{A2B1}	O
R	X_{A2B2}	O

- ▶ Three major advantages:
 - ▶ They often require fewer units.
 - ▶ They allow testing combinations of treatments more easily.
 - ▶ They allow testing interactions.

Example: Typing speed = $f(\text{Experience, Device})$

Part 1: Analyzing Dataset a

Initial Descriptives:

- ▶ Calculate the overall average typing speed for Novice users and Experienced users, ignoring the device.
- ▶ Then, calculate the overall average typing speed for the Mouse and the Touchscreen, ignoring experience.
- ▶ What do these marginal means suggest about the effects of experience and device type?

Part 1: Analyzing Dataset a

Main Effects Model:

- ▶ Run an ANOVA or linear regression predicting typing speed using only the main effects of Experience and Device (do not include an interaction term).
- ▶ Look at your p-values and R-squared. What is your conclusion so far?

Part 1: Analyzing Dataset a

Data Visualization:

- ▶ Create a line plot or interaction plot with Experience on the x-axis, Typing Speed on the y-axis, and separate lines for the Device.

Part 1: Analyzing Dataset a

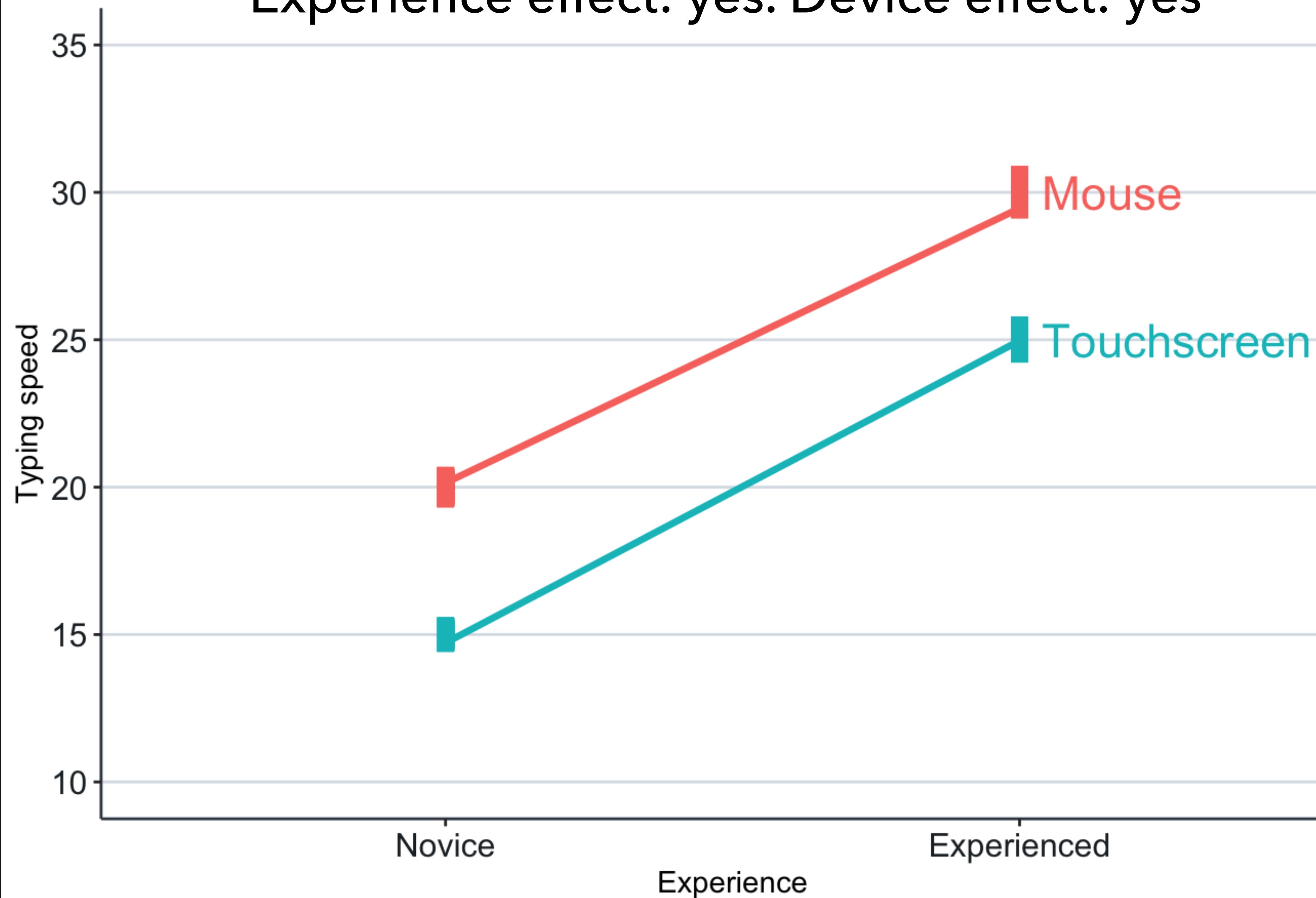
Full Factorial:

- ▶ Now, run the full factorial model (including the interaction term). Compare this to your initial conclusions.

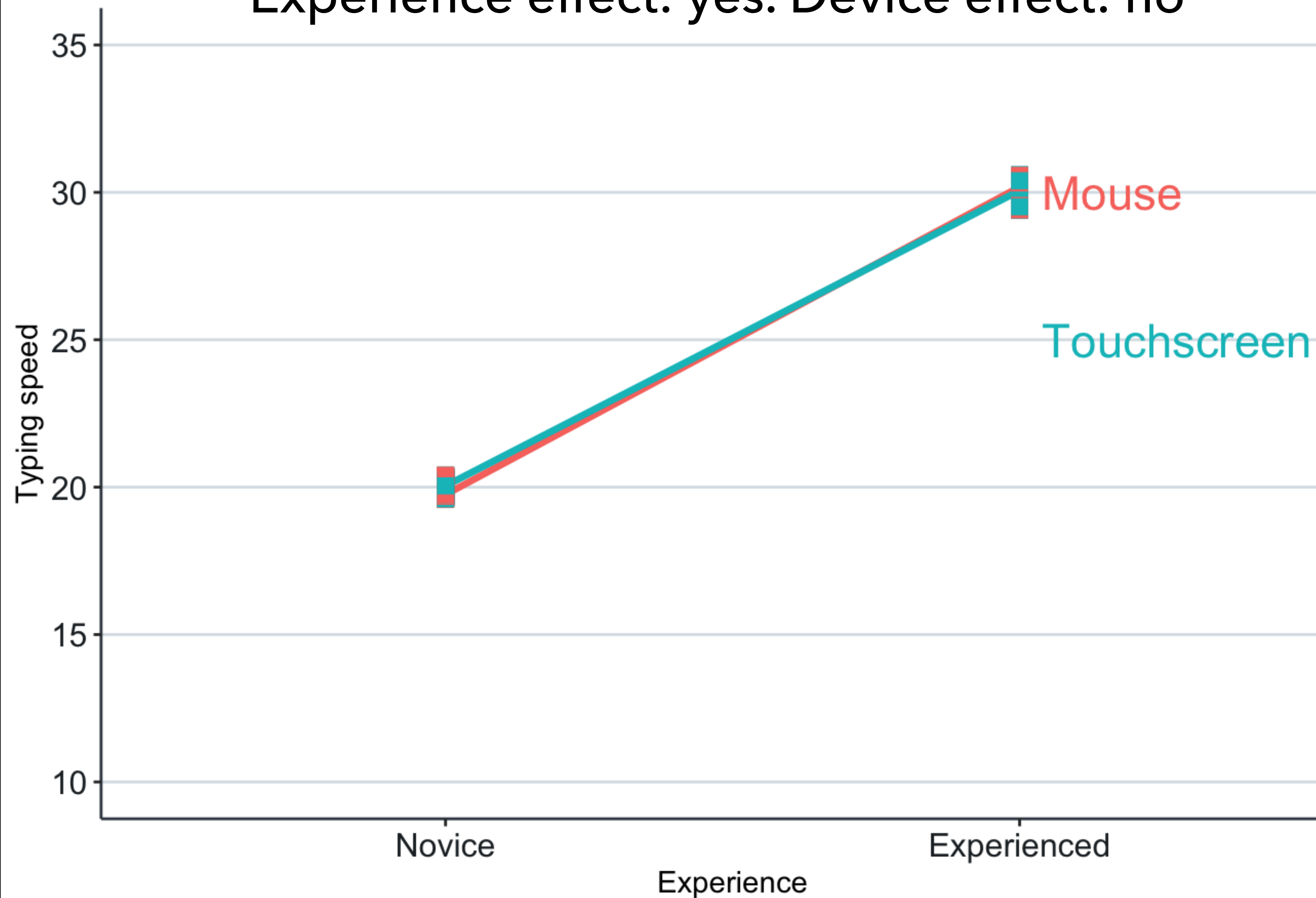
Part 2: Analyzing Dataset B

- ▶ Full Model: Run a full factorial ANOVA/regression (including main effects and the interaction) on Dataset B.
- ▶ Interpretation: Which effects are statistically significant?
- ▶ Look at the main effect of "Device" in your output. What does it suggest?
- ▶ Plot the data. Based on the plot, is it accurate to conclude that "Device X is faster than Device Y"? Why or why not?
- ▶ Write a short, two-sentence conclusion summarizing the findings of Dataset B as if you were writing the abstract of a paper. Be sure to accurately describe how experience changes the relationship between device and typing speed.

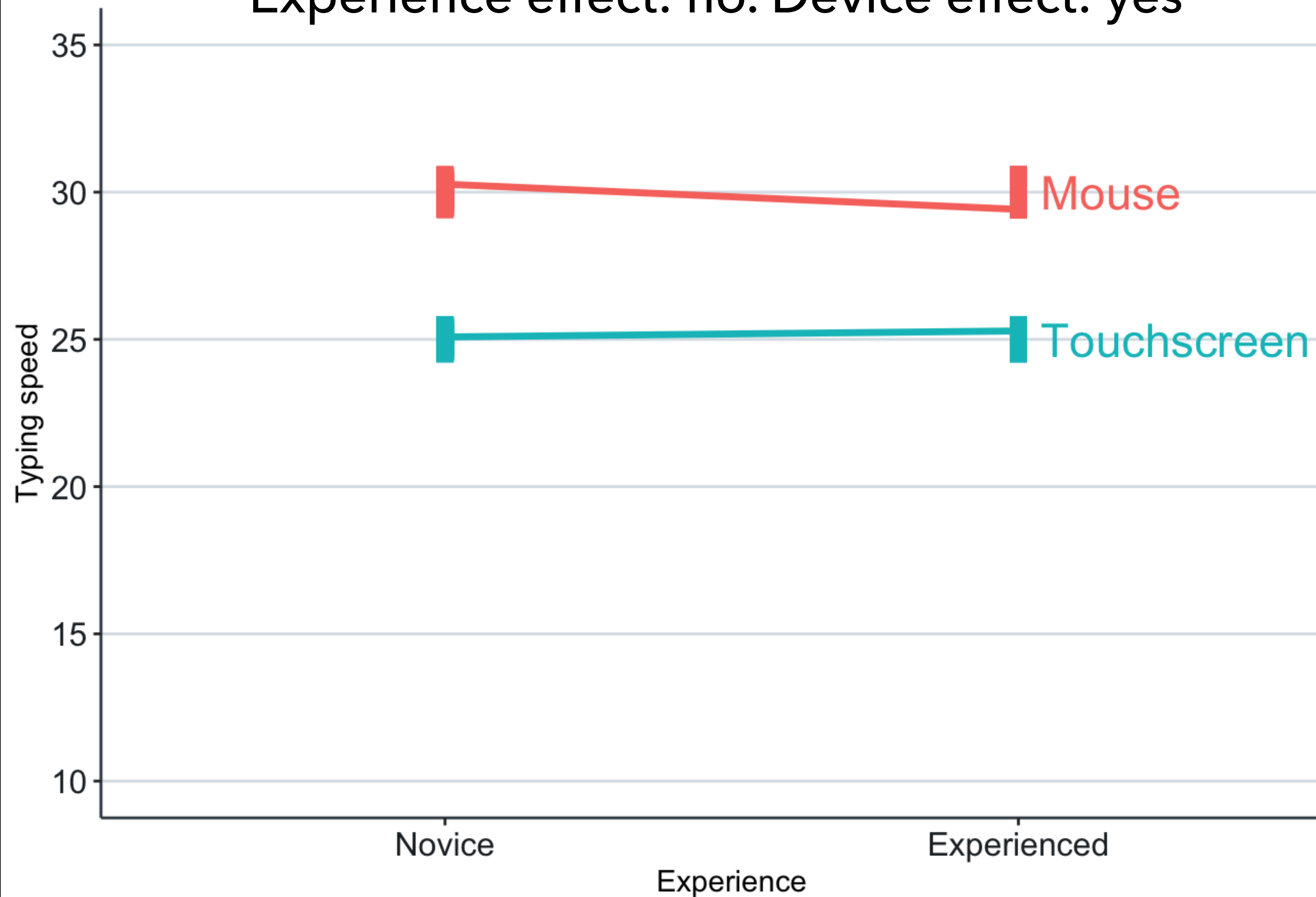
Experience effect: yes. Device effect: yes



Experience effect: yes. Device effect: no

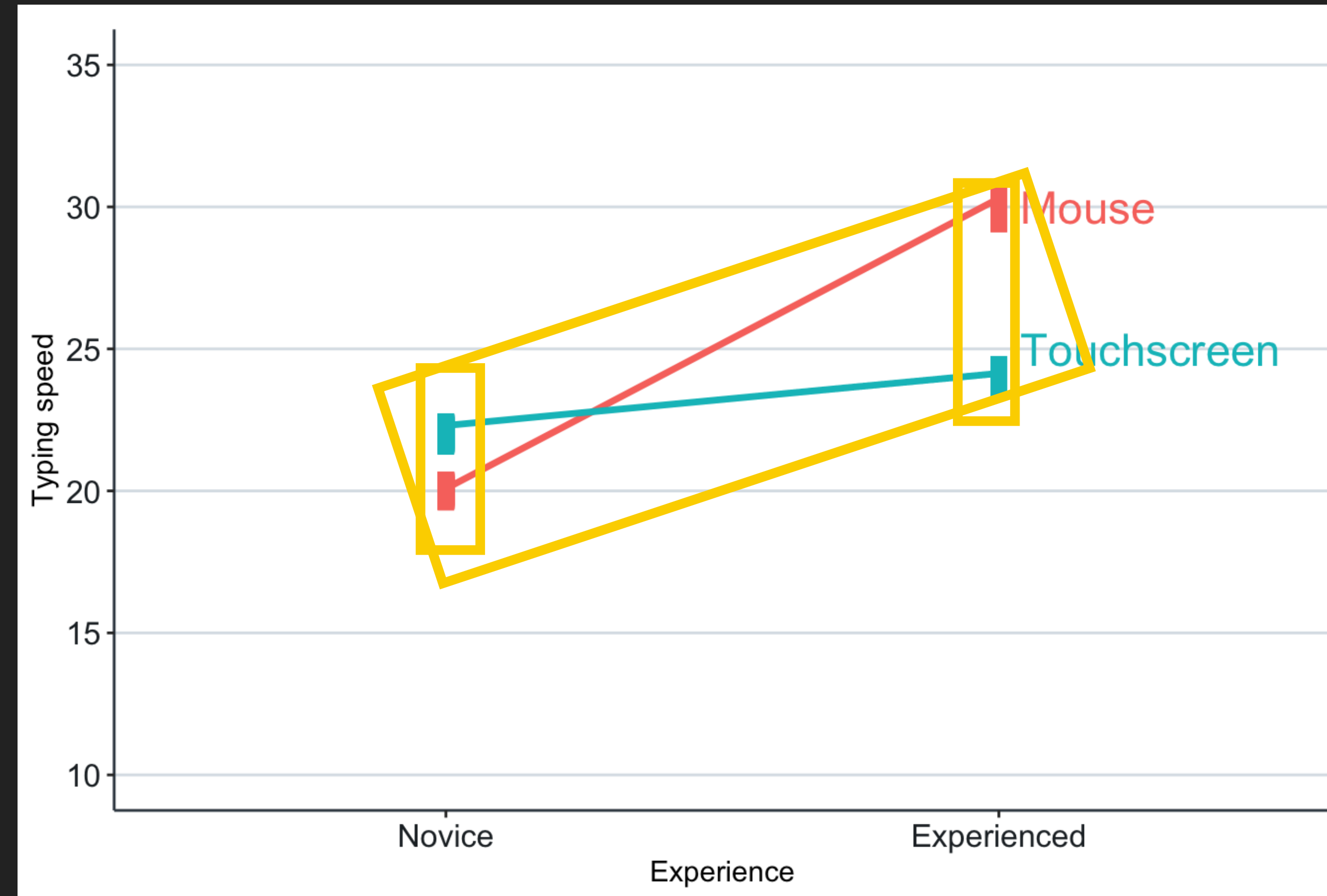


Experience effect: no. Device effect: yes



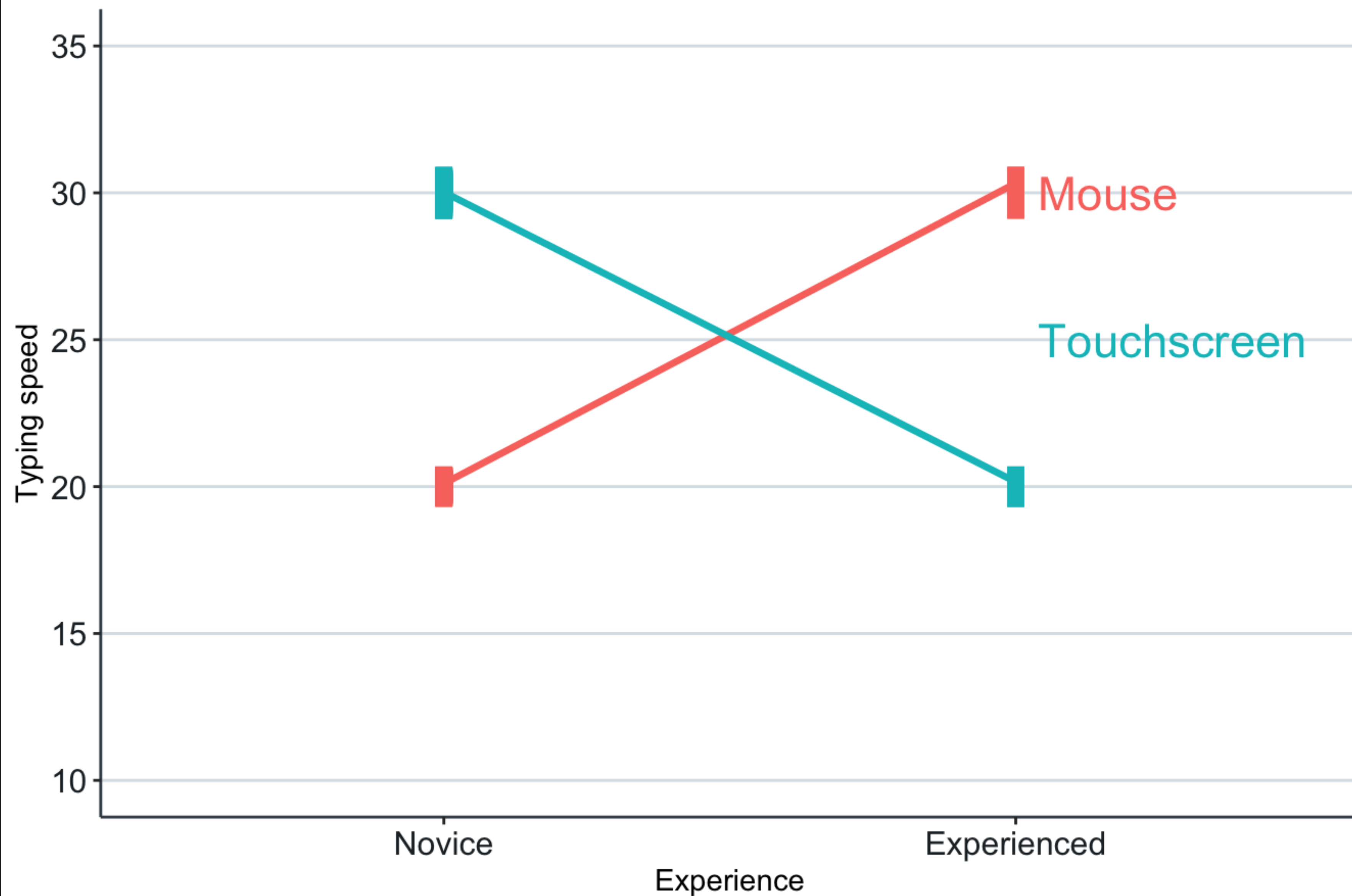
Example of Interaction Effects

- ▶ Novice users can select targets faster with a touchscreen than with a mouse.
- ▶ Experienced users can select targets faster with a mouse than with a touchscreen.
- ▶ The target selection speeds for both the mouse and the touchscreen increase as the user gains more experience with the device.
- ▶ However, the increase in speed is much larger for the mouse than for the touchscreen.





Experience effect: no. Device effect: no. Interaction: yes



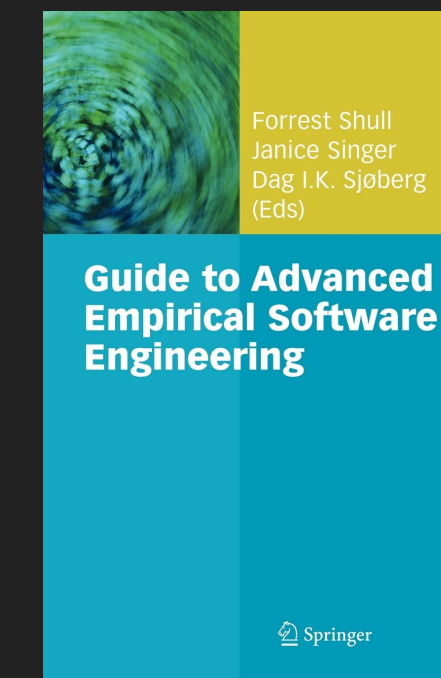
Credits

- ▶ Graphics: Dave DiCello photography (cover)
- ▶ Chapters from Shadish, W. R., Cook, T. D., & Campbell, D. T. (2002). Experimental and quasi-experimental designs for generalized causal inference. Wadsworth Publishing
 - ▶ Ch1: Experiments and generalized causal inference
 - ▶ Ch2: Statistical conclusion validity and internal validity
 - ▶ Ch3: Construct validity and external validity
 - ▶ Ch8: Randomized experiments
- ▶ Bruce, P., Bruce, A., & Gedeck, P. (2020). Practical Statistics for Data Scientists: 50+ Essential Concepts Using R and Python. O'Reilly Media.
- ▶ Freedman, D., Pisani, R., Purves, R., & Adhikari, A. (2007). Statistics.
- ▶ Goodman, S. (2008). A dirty dozen: Twelve p-value misconceptions. In Seminars in Hematology (Vol. 45, No. 3, pp. 135-140). WB Saunders.
- ▶ Lazar, J., Feng, J. H., & Hochheiser, H. (2017). Research methods in human-computer interaction. Morgan Kaufmann.
 - ▶ Ch 3: Experimental design
 - ▶ Ch 4: Statistical analysis
- ▶ MacKenzie, I. S. (2012). Human-computer interaction: An empirical research perspective.
 - ▶ Ch 6: Hypothesis testing
- ▶ Robertson, J., & Kaptein, M. (Eds.). (2016). Modern statistical methods for HCI. Cham: Springer.
 - ▶ Ch 5: Effect sizes and power analysis
 - ▶ Ch 13: Fair statistical communication
 - ▶ Ch 14: Improving statistical practice
- ▶ Kaptein, M., & Robertson, J. (2012). Rethinking statistical analysis methods for CHI. In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (pp. 1105-1114).

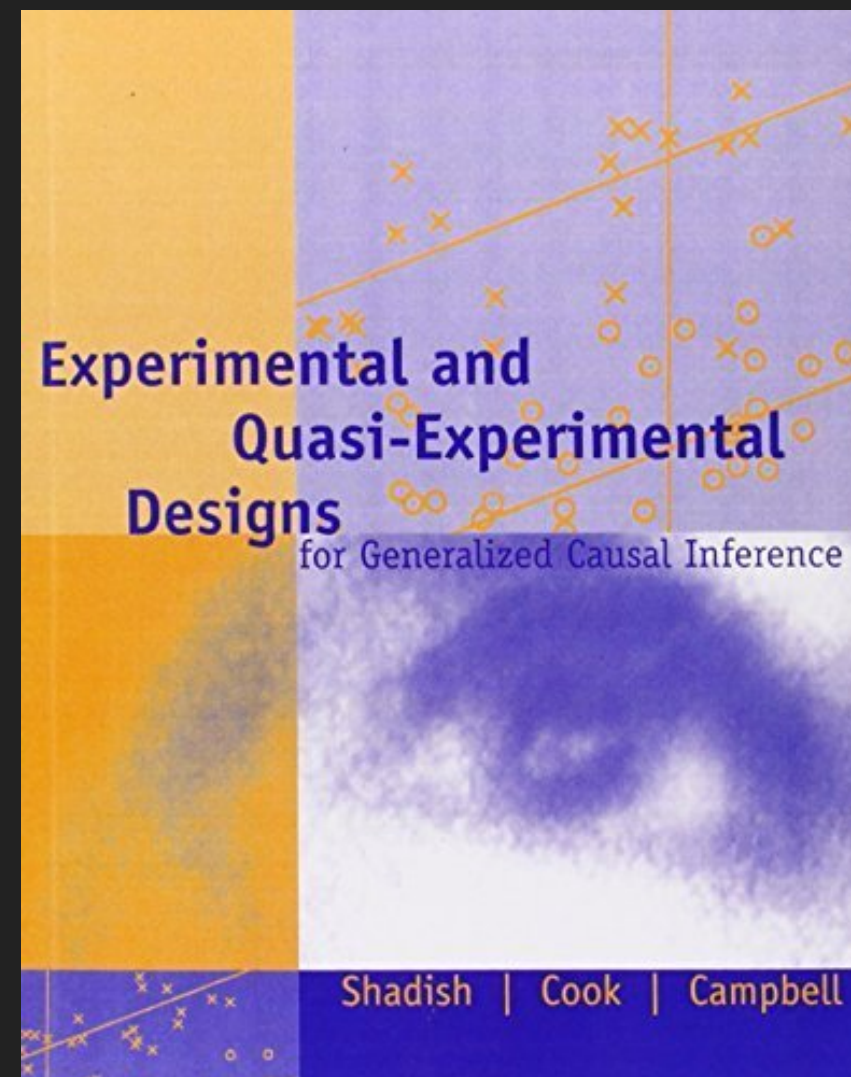
Read



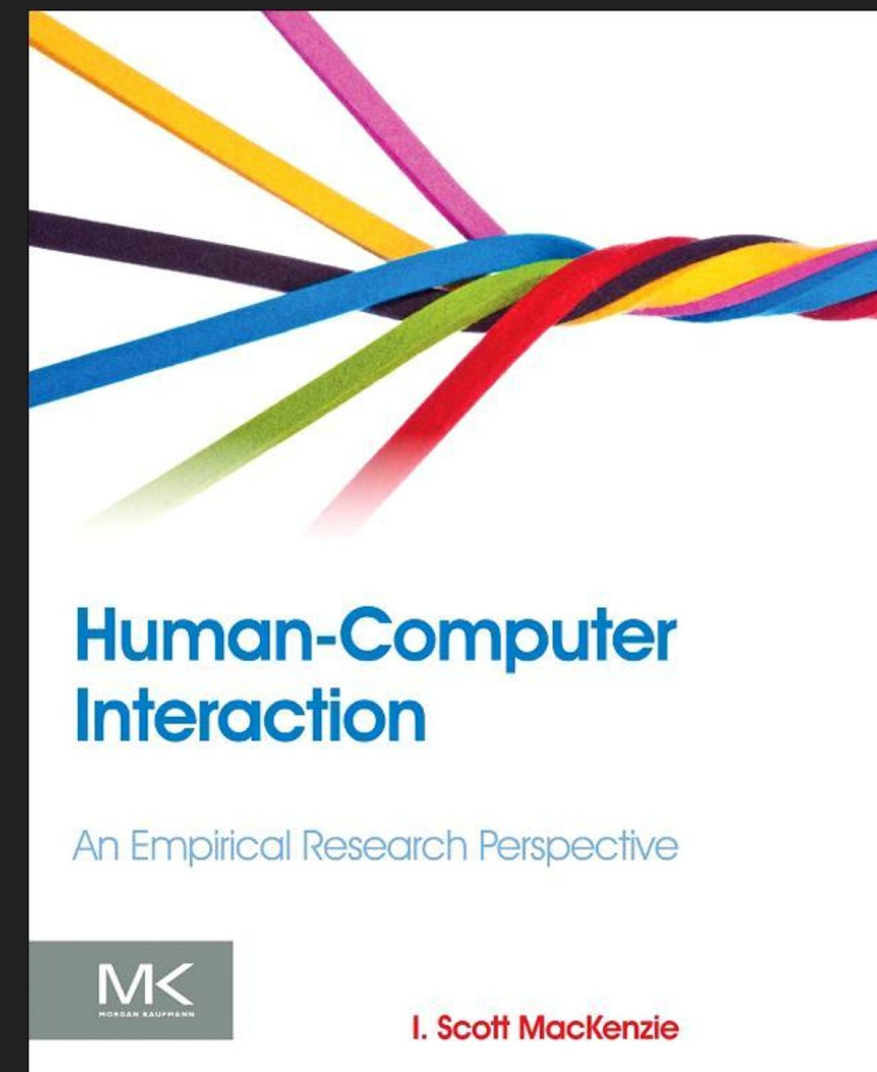
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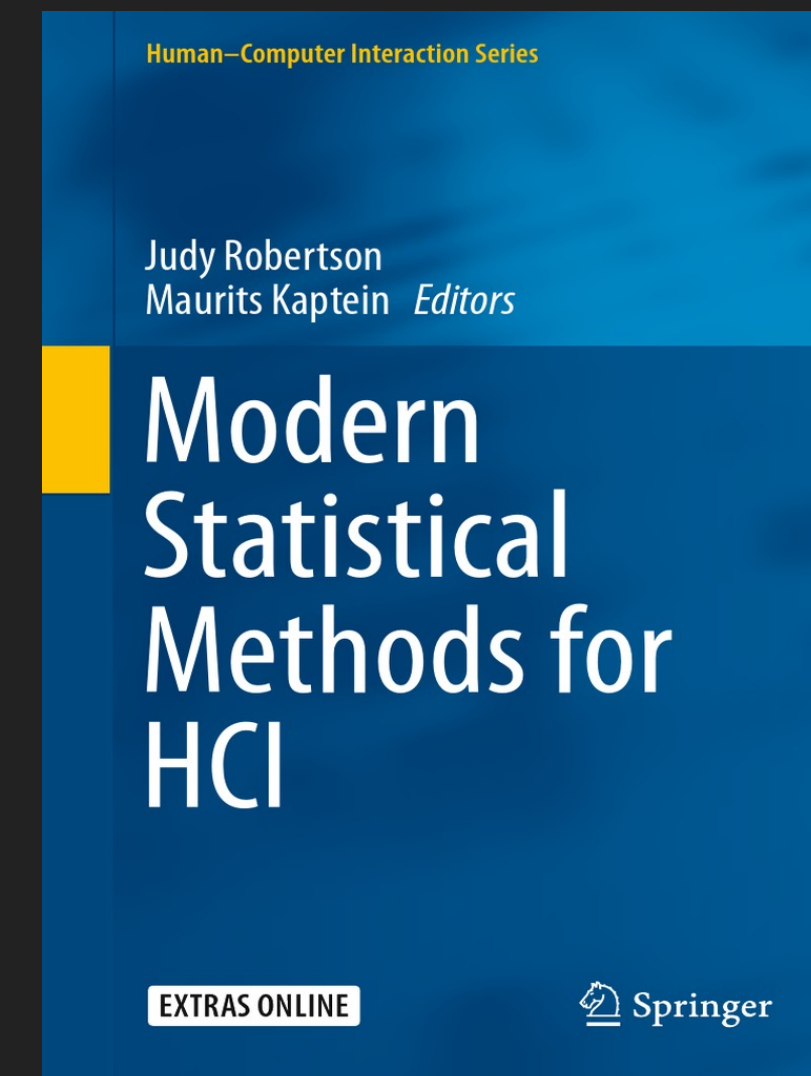
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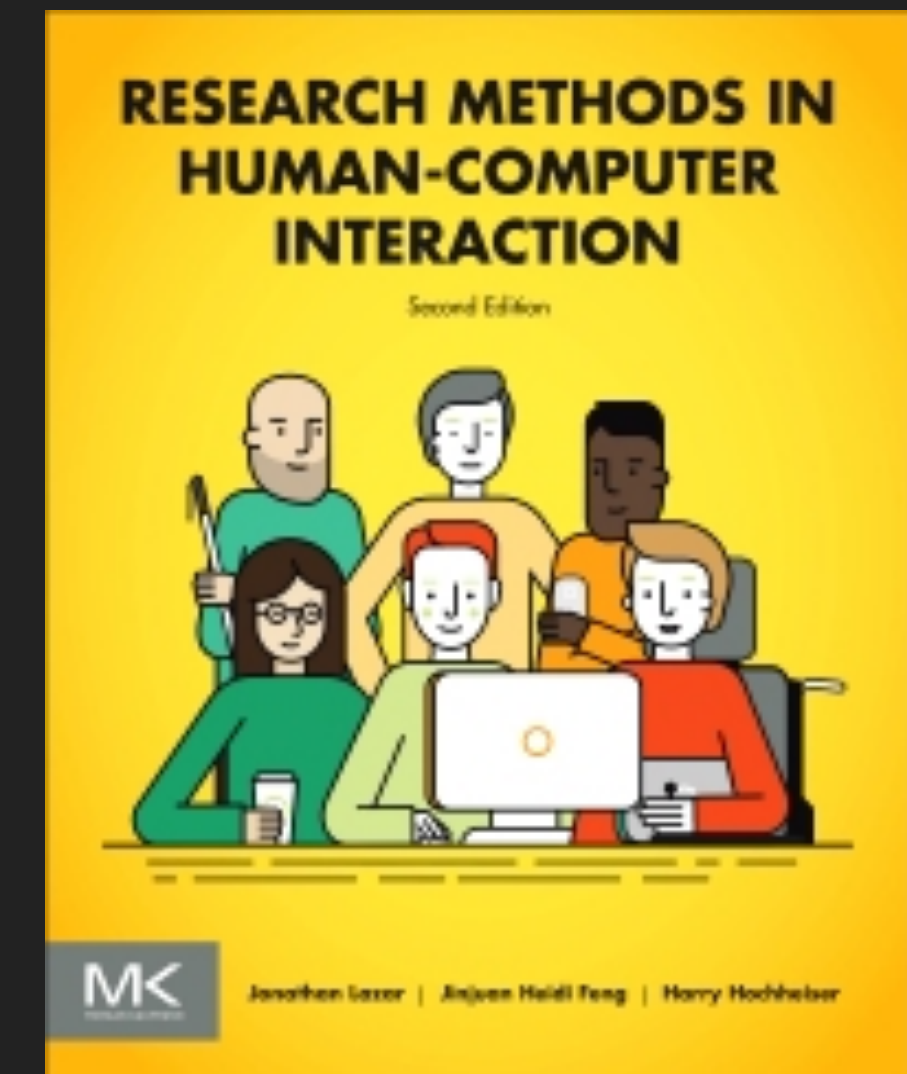
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